

Lifecycle Policies and cost optimization

Tasks-1. Create a new S3 bucket named 'my-zomato-images' and upload at least 3 sample image files representing restaurant menu items.

? **Click Create bucket.**

? **Enter:**

- **Bucket name: my-zomato-images**
- **Region: choose the region your assignment requires, e.g. Asia Pacific (Mumbai) – ap-south-1.**

? **Keep Block all public access enabled.**

? **Click Create bucket.**

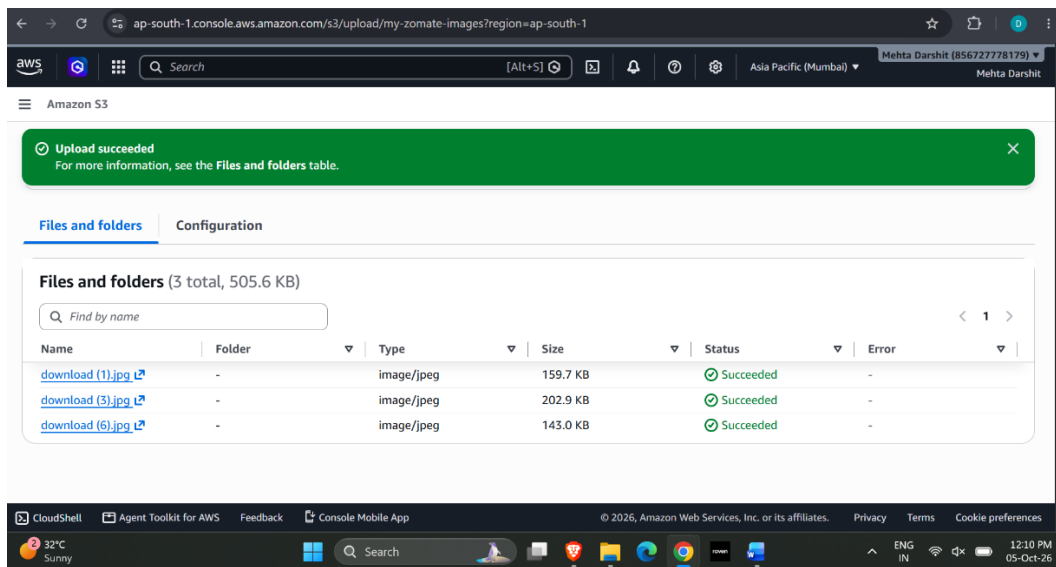
? **Open my-zomato-images.**

? **Click Upload → Add files.**

? **Select at least 3 sample restaurant/menu images, for example:**

- **pizza.jpg**
- **biryani.jpg**
- **burger.jpg**

? **Click Upload.**



Tasks-2. Set up a lifecycle policy on your 'my-zomato-images' S3 bucket to automatically transition all objects older than 30 days to the Glacier storage class. Use the S3 Management Console's 'Lifecycle rules' section to define the transition action.

Open:

S3 → my-zomato-images → Management → Lifecycle rules → Create lifecycle rule

Configure it like this:

Rule name:

move-images-to-glacier

Rule scope:

Select **Apply to all objects in the bucket.**

Confirm the rule applies to all objects.

Under **Lifecycle rule actions**, select:

☒ **Move current versions of objects between storage classes**

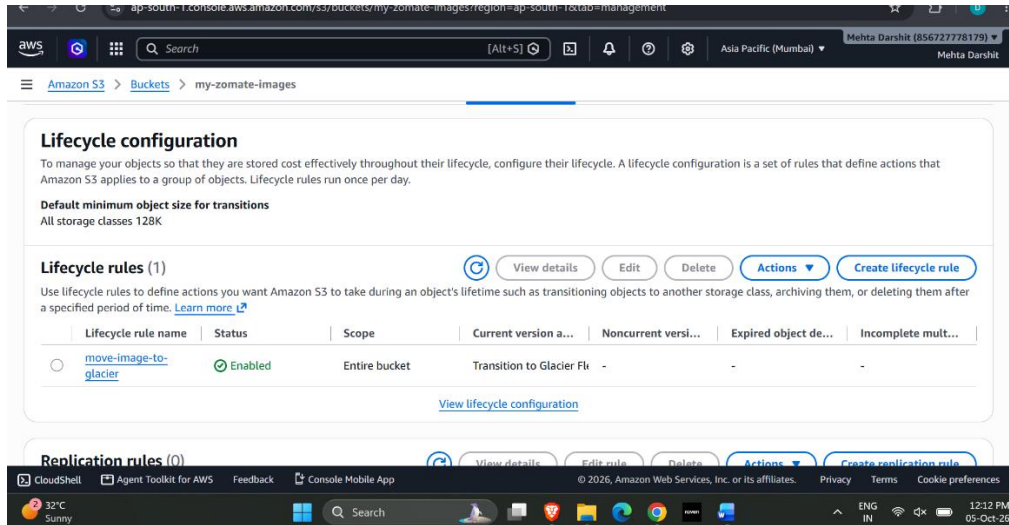
Then configure:

Transition to Glacier Flexible Retrieval

Days after object creation: 30

Then click **Create rule.**

AWS may display slightly different wording depending on the S3 console version.



Tasks-3. Simulate a cost optimization scenario: calculate and compare the estimated monthly storage cost for keeping 5GB of images in S3 Standard versus S3 Glacier using AWS Pricing Calculator. Use the AWS Pricing Calculator website and note down the difference in INR.

My Estimate								Duplicate Delete Move to Create group Add support Add service	
Find resources								< 1 > ⚙️	
☐	Service Name		Status	Upfront cost	Monthly cost	Description	Region		
☐	Amazon Simple Storage Service (S3)		-	0.00 USD	0.14 USD	standard	Asia Pacific (Mu...		
☐	Amazon Simple Storage Service (S3)		-	0.00 USD	0.26 USD	Glacier	Asia Pacific (Mu...		

[illegible]

S3 Standard	5 GB	₹13.4
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S3 Glacier	5 GB	₹25
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Difference **₹12.4**

Tasks-4.Add a new rule to your S3 bucket's lifecycle policy so that any objects with the prefix 'offers/' (e.g., 'offers/summer2024.jpg') are deleted automatically after 60 days.

Go back to:

S3 → my-zomato-images → Management → Lifecycle rules → Create lifecycle rule

Give it a name such as:

delete-old-offers

Choose the scope

Select:

Limit the scope to a prefix

Enter:

offers/

This means the rule applies to objects such as:

offers/summer2024.jpg

offers/diwali.jpg

offers/weekend-offer.png

but **not**:

pizza.jpg

burger.jpg

Lifecycle action

Select:

☒ **Permanently delete noncurrent versions of objects** / the appropriate object-expiration option shown by your console.

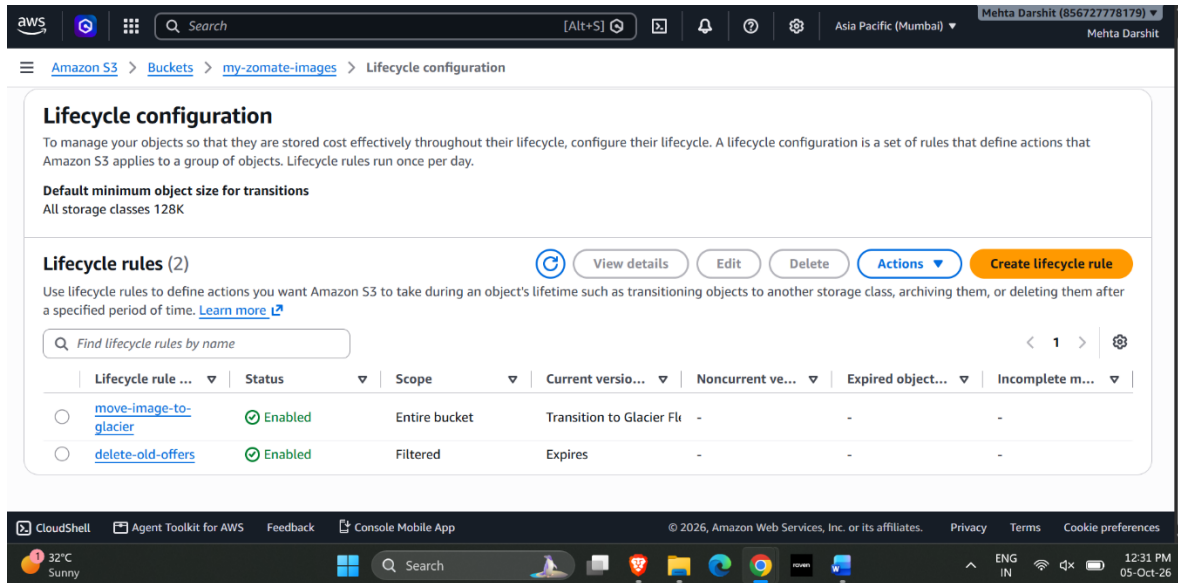
For the current objects, select:

☒ **Expire current versions of objects**

Set:

Days after object creation: 60

Then click **Create rule**.



Tasks-5. Use ChatGPT to generate a sample JSON configuration for an S3 lifecycle policy that transitions files to Glacier after 15 days and deletes them after 180 days. Paste the generated JSON and briefly explain each section.

```
{
  "Rules": [
    {
      "ID": "TransitionToGlacierAndDelete",
      "Status": "Enabled",
      "Filter": {},
      "Transitions": [
        {
          "Days": 15,
          "StorageClass": "GLACIER"
        }
      ],
    },
  ],
}
```

```
"Expiration": {  
  "Days": 180  
}  
}  
]  
}
```

Explanation

Rules

Contains the lifecycle rules that AWS should apply to objects.

ID

"ID": "TransitionToGlacierAndDelete"

A descriptive name identifying this lifecycle rule.

Status

"Status": "Enabled"

Activates the lifecycle rule.

Filter

"Filter": {}

An empty filter means the rule applies to all objects covered by the rule rather than only a particular prefix or tag.

Transitions

```
"Transitions": [  
  {  
    "Days": 15,  
    "StorageClass": "GLACIER"  
  }  
]
```

]

This tells S3 to transition objects to the **S3 Glacier Flexible Retrieval** storage class after **15 days**.

Expiration

"Expiration": {

 "Days": 180

}

This tells S3 to expire/delete the objects after **180 days**.